JAVA ASSIGNMENT

AIM:

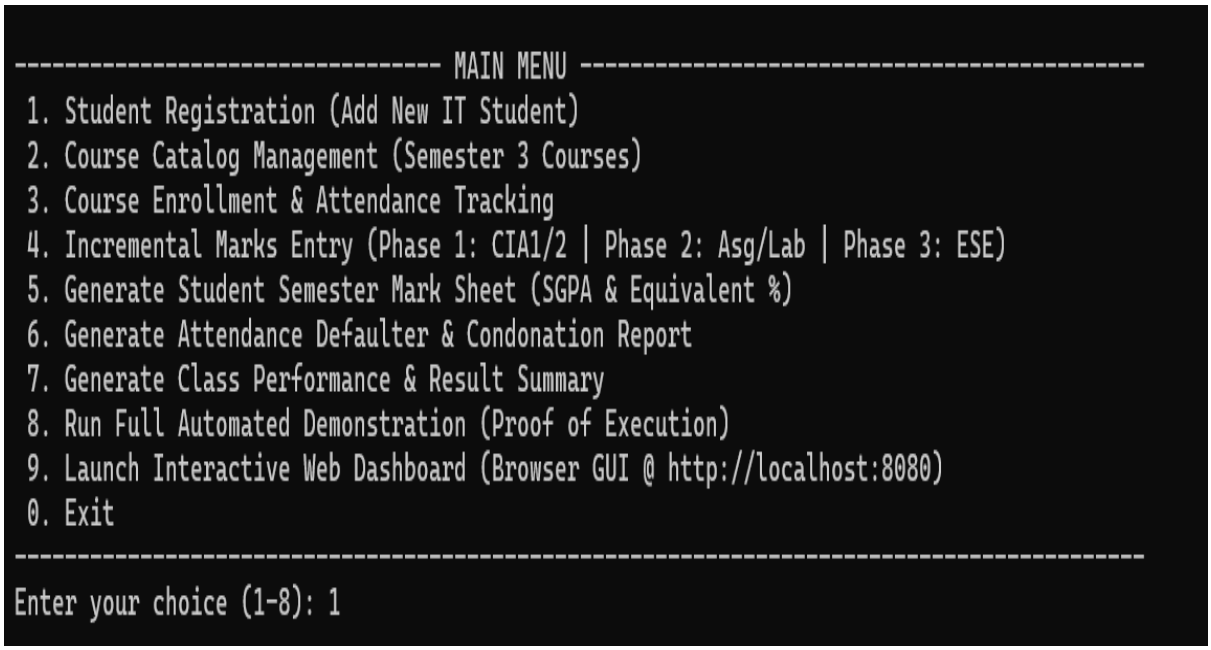
Development of Kongu Engineering College (R2024) Student Academic & Continuous Evaluation System using Java JDBC.

OBJECTIVE:

Design and develop a database-driven Java application using JDBC to manage student records, perform incremental marks entry (Continuous Internal Assessments and End Semester Examinations), enforce attendance eligibility rules, and generate formal academic performance reports according to

Kongu Engineering College (Autonomous) - Regulation 2024. System Requirements & Specifications.

MAIN MENU:



STUDENT & COURSE MANAGEMENT:

1.STUDENT REGISTRATION:

```
--- [1] Student Registration ---
Enter Roll Number (e.g., 24ITR050): 25ITL187
Enter Student Full Name: Lingesh B
Enter Department [Press Enter for IT]:
Enter Current Semester [Press Enter for 3]:
[SUCCESS] Student 25ITL187 (Lingesh B) registered successfully.

Press [Enter] to return to the Main Menu...
```

2. COURSE CATALOG:

```
--- [2] Course Catalog Management ---
Enter Course Code (e.g., 24ITR31 / 24ITT32): 24ITT11
Enter Course Title: Game Development
Enter Credits (e.g., 2, 3, 4): 2
Enter Course Type (1: THEORY, 2: PRACTICAL, 3: EMBEDDED): 2
Enter Offering Semester (1-8): 4
[SUCCESS] Course 24ITT11 [PRACTICAL, 2 Credits] added successfully.

Press [Enter] to return to the Main Menu...
```

3. COURSE ENROLLMENT & ATTENDANCE TRACKING:

```
--- [3] Course Enrollment & Attendance Tracking ---
Enter Student Roll Number: 25ITL187
Enter Course Code: 24ITC31
Enter Semester: 3
Enter Total Classes Held: 60
Enter Classes Attended: 52
Calculated Attendance: 86.67%
[INFO] Attendance >= 75%. Eligible for ESE.
[SUCCESS] Enrollment and Attendance updated. (Enrollment ID: 413)

Press [Enter] to return to the Main Menu...
```

INCREMENTAL MARKS ENTRY:

1.PHASE 1 (MID-SEMESTER):

```
----------------- INCREMENTAL MARKS ENTRY (THEN AND NOW) -----------------
 Select Assessment Phase:
 1. Phase 1 (Mid-Semester): CIA Test 1 (0-100) & CIA Test 2 (0-100)
 2. Phase 2 (End of Semester - CIA): Assignments / Quizzes / Lab (0-100)
 3. Phase 3 (Post Semester): End Semester Exam (0-100) & Absentee (AB)
Enter Phase (1-3): 1
Enter Student Roll Number: 25ITL187
Enter Course Code: 24ITC31
Enter Semester: 3
Enter CIA Test 1 Marks (0-100): 92
Enter CIA Test 2 Marks (0-100): 93
[SUCCESS] Phase 1 Marks recorded successfully.

Press [Enter] to return to the Main Menu...
```

2.PHASE 2 (END OF SEMESTER - CIA):

```
----------------- INCREMENTAL MARKS ENTRY (THEN AND NOW) -----------------
 Select Assessment Phase:
 1. Phase 1 (Mid-Semester): CIA Test 1 (0-100) & CIA Test 2 (0-100)
 2. Phase 2 (End of Semester - CIA): Assignments / Quizzes / Lab (0-100)
 3. Phase 3 (Post Semester): End Semester Exam (0-100) & Absentee (AB)
Enter Phase (1-3): 2
Enter Student Roll Number: 25ITL187
Enter Course Code: 24ITC31
Enter Semester: 3
Enter Assignment / Quiz / Lab Evaluation Marks (0-100): 90
[SUCCESS] Phase 2 Marks recorded successfully.

Press [Enter] to return to the Main Menu...
```

3. PHASE 3 (POST SEMESTER):

```
----------------- INCREMENTAL MARKS ENTRY (THEN AND NOW) -----------------
 Select Assessment Phase:
 1. Phase 1 (Mid-Semester): CIA Test 1 (0-100) & CIA Test 2 (0-100)
 2. Phase 2 (End of Semester - CIA): Assignments / Quizzes / Lab (0-100)
 3. Phase 3 (Post Semester): End Semester Exam (0-100) & Absentee (AB)
Enter Phase (1-3): 3
Enter Student Roll Number: 25ITL187
Enter Course Code: 24ITC31
Enter Semester: 3
Is student Absent (AB) in ESE? (y/n): n
Enter End Semester Exam (ESE) Marks (0-100): 87
[SUCCESS] Phase 3 ESE Marks recorded successfully.

Press [Enter] to return to the Main Menu...
```

ATTENDANCE & GRADING RULES:

1.ATTENDANCE FILTER:

```
Enter your choice (1-8): 6

Enter Semester to check (e.g., 3): 3

=============================================================================================================
                      KONGU ENGINEERING COLLEGE (AUTONOMOUS) - REGULATION 2024
                           ATTENDANCE DEFAULTER & CONDONATION REPORT
                                      SEMESTER: 3
=============================================================================================================
Roll No      Student Name         Dept     Course     Held     Attended Attd %     Condonation  Eligibility / Grade
-------------------------------------------------------------------------------------------------------------
24ADR010     Ananya S             AIDS     24ADT31    60       32          53.3% NOT APPROVED BARRED (Grade SA)
24ADR010     Ananya S             AIDS     24ADC31    60       32          53.3% NOT APPROVED BARRED (Grade SA)
24ADR010     Ananya S             AIDS     24ADL31    30       16          53.3% NOT APPROVED BARRED (Grade SA)
24CER003     Aravind K            CIVIL    24CET31    60       42          70.0% APPROVED     CONDONED (Eligible)
24CER003     Aravind K            CIVIL    24CEC31    60       42          70.0% APPROVED     CONDONED (Eligible)
24CER003     Aravind K            CIVIL    24CEL31    30       21          70.0% NOT APPROVED PENDING (Grade SA)
-------------------------------------------------------------------------------------------------------------
 Note: <65% Attendance: Shortage of Attendance (SA - Barred) | 65-74.9%: Requires Medical Condonation
=============================================================================================================


Press [Enter] to return to the Main Menu...
```

STUDENT SEMESTER MARK SHEET:

```
Enter Student Roll Number (e.g., 24ITR001): 24ITR099
Enter Semester (e.g., 3): 3

=================================================================================================================
                                    KONGU ENGINEERING COLLEGE (AUTONOMOUS)
                                        PERUNDURAI, ERODE - 638060
                                  OFFICE OF THE CONTROLLER OF EXAMINATIONS
                               GRADE SHEET - REGULATION 2024 (AUTONOMOUS)
=================================================================================================================
 Roll Number : 24ITR099          Student Name : Murasoli B                      Semester : 3
 Department  : IT                Regulation   : R2024 (KEC)                      Date     : 20-10-2026
-----------------------------------------------------------------------------------------------------------------
Course    Course Title               Type      Cr   Attd%    CIA1   CIA2   Asg    Sc.CIA   ESE(100) Sc.ESE  Total  Grd  Result
-----------------------------------------------------------------------------------------------------------------
24ITC31   Full Stack Development      EMBEDDED  4      96.7% 94.0   96.0   95.0      47.50 94.0     47.0     94.50 O    PASS
24ITL31   Programming in Java Labo...  PRACTICAL 1     100.0% 98.0   98.0   100.0     59.16 96.0     38.4     97.56 O    PASS
24ITT32   Programming in Java         THEORY    3      97.8% 90.0   92.0   94.0      36.76 90.0     54.0     90.76 O    PASS
-----------------------------------------------------------------------------------------------------------------
 Total Enrolled Credits : 8      |   Credits Earned : 8      |   Weighted Grade Points : 80.00
 SEMESTER GRADE POINT AVERAGE (SGPA) : 10.00 / 10.00
 EQUIVALENT PERCENTAGE               : 100.00%
 SEMESTER EVALUATION OUTCOME         : [PASSED] ALL REGISTERED COURSES CLEARED
=================================================================================================================
 Grading Standard (R2024): O(91-100)->10 | A+(81-90)->9 | A(71-80)->8 | B+(61-70)->7 | B(50-60)->6 | RA/SA/AB->0
 Rules: Theory (40 CIA/60 ESE, ESE Min 27/60) | Practical (60 CIA/40 ESE, ESE Min 20/40) | Pass Minimum: 50 Marks
=================================================================================================================


Press [Enter] to return to the Main Menu...
```

DEPARTMENT WISE CLASS PERFORMANCE & RESULT SUMMARY:

```
Enter your choice (1-8): 7

Enter Semester (e.g., 3): 3

================================================================================
                    KONGU ENGINEERING COLLEGE (AUTONOMOUS) - REGULATION 2024
                     DEPARTMENT-WISE CLASS PERFORMANCE & RESULT SUMMARY
                                      SEMESTER: 3
================================================================================
Department   Total Students   Passed All     With Arrears   Pass Rate (%)  Class Avg SGPA
--------------------------------------------------------------------------------
AIDS         1               0              1                     0.00%           0.00
CHEM         1               1              0                   100.00%           9.11
CIVIL        1               0              1                     0.00%           8.44
CSE          2               2              0                   100.00%           8.42
ECE          1               1              0                   100.00%           9.00
EEE          1               1              0                   100.00%           9.00
IT           5               5              0                   100.00%           8.47
MECH         1               0              1                     0.00%           6.58
================================================================================


Press [Enter] to return to the Main Menu...
```

DATABASE:

STUDENTS DATABASE:

```
mysql> USE kec_eval;
Database changed
mysql> SELECT * FROM students;
+----------+-------------------+------------+------------------+---------------------+
| roll_no  | name              | department | current_semester | created_at          |
+----------+-------------------+------------+------------------+---------------------+
| 22ADR071 | Sudharsan G       | AIDS       |                7 | 2026-10-19 21:29:39 |
| 22ADR081 | Akilesh R         | AIDS       |                8 | 2026-10-19 21:29:39 |
| 22CER071 | Saranya V         | CIVIL      |                7 | 2026-10-19 21:29:39 |
| 22CER081 | Yogeshwaran T     | CIVIL      |                8 | 2026-10-19 21:29:39 |
| 22CHR071 | Swetha T          | CHEM       |                7 | 2026-10-19 21:29:39 |
| 22CHR081 | Balamurugan S     | CHEM       |                8 | 2026-10-19 21:29:39 |
| 22CSR071 | Praveen Kumar R   | CSE        |                7 | 2026-10-19 21:29:39 |
| 22CSR072 | Priyanga M        | CSE        |                7 | 2026-10-19 21:29:39 |
| 22CSR081 | Venkatesh S       | CSE        |                8 | 2026-10-19 21:29:39 |
| 22CSR082 | Vidhya R          | CSE        |                8 | 2026-10-19 21:29:39 |
| 22ECR071 | Rajesh K          | ECE        |                7 | 2026-10-19 21:29:39 |
| 22ECR081 | Vigneshwar B      | ECE        |                8 | 2026-10-19 21:29:39 |
| 22EER071 | Rithika S         | EEE        |                7 | 2026-10-19 21:29:39 |
| 22EER081 | Vishnu Vardhan M  | EEE        |                8 | 2026-10-19 21:29:39 |
| 22ITR071 | Manikandan P      | IT         |                7 | 2026-10-19 21:29:39 |
| 22ITR072 | Nandhini S        | IT         |                7 | 2026-10-19 21:29:39 |
```

NAME: MURASOLI B ROLL NO: 25ITL188

COURSES DATABASE:

```
mysql> SELECT * FROM courses;
+-------------+-----------------------------------------------+---------+-------------+----------+---------------------+
| course_code | title                                         | credits | course_type | semester | created_at          |
+-------------+-----------------------------------------------+---------+-------------+----------+---------------------+
| 24ADC31     | Artificial Intelligence & Knowledge Rep.      |       4 | EMBEDDED    |        3 | 2026-10-19 20:59:10 |
| 24ADC41     | Machine Learning Algorithms & Systems         |       4 | EMBEDDED    |        4 | 2026-10-19 20:59:10 |
| 24ADC51     | Natural Language Processing (NLP)             |       4 | EMBEDDED    |        5 | 2026-10-19 20:59:10 |
| 24ADC61     | Generative AI & Large Language Models         |       4 | EMBEDDED    |        6 | 2026-10-19 20:59:10 |
| 24ADC71     | Reinforcement Learning & Autonomous Agents    |       4 | EMBEDDED    |        7 | 2026-10-19 20:59:10 |
| 24ADL31     | Python for AI & Data Science Laboratory       |       1 | PRACTICAL   |        3 | 2026-10-19 20:59:10 |
| 24ADL41     | Machine Learning & Data Mining Lab            |       1 | PRACTICAL   |        4 | 2026-10-19 20:59:10 |
| 24ADL51     | Deep Learning & NLP Laboratory                |       1 | PRACTICAL   |        5 | 2026-10-19 20:59:10 |
| 24ADL61     | Computer Vision & GenAI Laboratory            |       1 | PRACTICAL   |        6 | 2026-10-19 20:59:10 |
| 24ADP71     | AI Capstone Project Phase I                   |       3 | PRACTICAL   |        7 | 2026-10-19 20:59:10 |
| 24ADP81     | AI Capstone Project Phase II / Industry Track |      10 | PRACTICAL   |        8 | 2026-10-19 20:59:10 |
| 24ADT31     | Mathematical Foundations for Data Science     |       4 | THEORY      |        3 | 2026-10-19 20:59:10 |
| 24ADT41     | Statistical Learning & Data Mining            |       4 | THEORY      |        4 | 2026-10-19 20:59:10 |
| 24ADT51     | Deep Learning & Neural Networks               |       4 | THEORY      |        5 | 2026-10-19 20:59:10 |
| 24ADT61     | Computer Vision & Video Analytics             |       4 | THEORY      |        6 | 2026-10-19 20:59:10 |
| 24ADT71     | AI Ethics, Governance & MLOps Infrastructure  |       3 | THEORY      |        7 | 2026-10-19 20:59:10 |
| 24CEC31     | Surveying and Geomatics                       |       4 | EMBEDDED    |        3 | 2026-10-19 20:59:10 |
| 24CEC41     | Fluid Mechanics & Hydraulic Machinery         |       4 | EMBEDDED    |        4 | 2026-10-19 20:59:10 |
| 24CEC51     | Soil Mechanics & Geotechnical Engineering     |       4 | EMBEDDED    |        5 | 2026-10-19 20:59:10 |
| 24CEC61     | Environmental Engineering & Water Treatment   |       4 | EMBEDDED    |        6 | 2026-10-19 20:59:10 |
| 24CEC71     | Transportation Engineering & GIS              |       4 | EMBEDDED    |        7 | 2026-10-19 20:59:10 |
| 24CEL31     | Surveying Field Work Laboratory               |       1 | PRACTICAL   |        3 | 2026-10-19 20:59:10 |
| 24CEL41     | Materials Testing & Hydraulics Lab            |       1 | PRACTICAL   |        4 | 2026-10-19 20:59:10 |
| 24CEL51     | Geotechnical Engineering Lab                  |       1 | PRACTICAL   |        5 | 2026-10-19 20:59:10 |
| 24CEL61     | Environmental Engineering Laboratory          |       1 | PRACTICAL   |        6 | 2026-10-19 20:59:10 |
| 24CEP71     | Project Work Phase I                          |       3 | PRACTICAL   |        7 | 2026-10-19 20:59:10 |
```

ENROLLMENTS DATABASE:

```
mysql> SELECT * FROM enrollments;
+---------------+----------+-------------+----------+---------------+------------------+------------+---------------------+
| enrollment_id | roll_no  | course_code | semester | total_classes | attended_classes | is_condoned | created_at          |
+---------------+----------+-------------+----------+---------------+------------------+------------+---------------------+
|             1 | 24ITR001 | 24ITT31     |        3 |            60 |               49 |          0 | 2026-10-19 20:59:10 |
|             2 | 24ITR001 | 24ITC31     |        3 |            60 |               49 |          0 | 2026-10-19 20:59:10 |
|             3 | 24ITR001 | 24ITT32     |        3 |            60 |               51 |          0 | 2026-10-19 20:59:10 |
|             4 | 24ITR001 | 24ITC32     |        3 |            60 |               51 |          0 | 2026-10-19 20:59:10 |
|             5 | 24ITR001 | 24ITT33     |        3 |            60 |               53 |          0 | 2026-10-19 20:59:10 |
|             6 | 24ITR001 | 24MNT31     |        3 |            60 |               55 |          0 | 2026-10-19 20:59:10 |
|             7 | 24ITR001 | 24ITL31     |        3 |            30 |               26 |          0 | 2026-10-19 20:59:10 |
|             8 | 24ITR001 | 24ITL32     |        3 |            30 |               27 |          0 | 2026-10-19 20:59:10 |
|             9 | 24ITR001 | 24ITL33     |        3 |            30 |               28 |          0 | 2026-10-19 20:59:10 |
|            10 | 24ITR002 | 24ITT31     |        3 |            60 |               51 |          0 | 2026-10-19 20:59:10 |
|            11 | 24ITR002 | 24ITC31     |        3 |            60 |               51 |          0 | 2026-10-19 20:59:10 |
|            12 | 24ITR002 | 24ITT32     |        3 |            60 |               53 |          0 | 2026-10-19 20:59:10 |
|            13 | 24ITR002 | 24ITC32     |        3 |            60 |               53 |          0 | 2026-10-19 20:59:10 |
|            14 | 24ITR002 | 24ITT33     |        3 |            60 |               55 |          0 | 2026-10-19 20:59:10 |
|            15 | 24ITR002 | 24MNT31     |        3 |            60 |               56 |          0 | 2026-10-19 20:59:10 |
|            16 | 24ITR002 | 24ITL31     |        3 |            30 |               27 |          0 | 2026-10-19 20:59:10 |
|            17 | 24ITR002 | 24ITL32     |        3 |            30 |               28 |          0 | 2026-10-19 20:59:10 |
|            18 | 24ITR002 | 24ITL33     |        3 |            30 |               25 |          0 | 2026-10-19 20:59:10 |
|            19 | 24ITR003 | 24ITT31     |        3 |            60 |               53 |          0 | 2026-10-19 20:59:10 |
|            20 | 24ITR003 | 24ITC31     |        3 |            60 |               53 |          0 | 2026-10-19 20:59:10 |
|            21 | 24ITR003 | 24ITT32     |        3 |            60 |               55 |          0 | 2026-10-19 20:59:10 |
|            22 | 24ITR003 | 24ITC32     |        3 |            60 |               55 |          0 | 2026-10-19 20:59:10 |
|            23 | 24ITR003 | 24ITT33     |        3 |            60 |               56 |          0 | 2026-10-19 20:59:10 |
|            24 | 24ITR003 | 24MNT31     |        3 |            60 |               49 |          0 | 2026-10-19 20:59:10 |
|            25 | 24ITR003 | 24ITL31     |        3 |            30 |               28 |          0 | 2026-10-19 20:59:10 |
|            26 | 24ITR003 | 24ITL32     |        3 |            30 |               25 |          0 | 2026-10-19 20:59:10 |
|            27 | 24ITR003 | 24ITL33     |        3 |            30 |               26 |          0 | 2026-10-19 20:59:10 |
|            28 | 24CSR001 | 24CST31     |        3 |            60 |               49 |          0 | 2026-10-19 20:59:10 |
|            29 | 24CSR001 | 24CSC31     |        3 |            60 |               55 |          0 | 2026-10-19 20:59:10 |
|            30 | 24CSR001 | 24CST32     |        3 |            60 |               51 |          0 | 2026-10-19 20:59:10 |
```

STUDENTS DECRIPTION:

```
mysql> DESC students;
+------------------+--------------+------+-----+-------------------+-------------------+
| Field            | Type         | Null | Key | Default           | Extra             |
+------------------+--------------+------+-----+-------------------+-------------------+
| roll_no          | varchar(20)  | NO   | PRI | NULL              |                   |
| name             | varchar(100) | NO   |     | NULL              |                   |
| department       | varchar(20)  | NO   |     | NULL              |                   |
| current_semester | int          | NO   |     | 3                 |                   |
| created_at       | timestamp    | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+------------------+--------------+------+-----+-------------------+-------------------+
5 rows in set (0.04 sec)
```

ALGORITHM / PROCEDURE:

The system initiates by establishing a secure JDBC connection between the Java application layer and the MySQL relational database (kec_eval) using the MySQL Connector/J driver. Upon user request, it queries the student profile, course catalog, attendance figures, and assessment scores using parameterized SQL queries through Data Access Objects (DAO). The algorithm first evaluates attendance eligibility for each registered course: students with attendance $\geq 75\%$qualify directly for examinations, those between 65%and 74%require verified medical condonation, and those below 65%are flagged with Shortage of Attendance (Grade SA) and barred from writing the End Semester Examination (ESE). Next, continuous internal assessment (CIA) and ESE marks are converted into normalized weightages according to the course type—Theory courses allocate 40% for CIA (comprising CIA-1, CIA-2, and Assignments) and 60% for ESE, whereas Practical courses allocate 60% for continuous lab evaluation and 40% for the practical examination. If the student satisfies both the mandatory ESE minimum and the overall pass threshold of 50 marks, the total score is mapped to the corresponding 10-point scale Letter Grade (O, A+, A, B+, B); otherwise, a Re-Appearance grade (RA) is awarded. Finally, the Semester Grade Point Average (SGPA) is computed by dividing the sum of credit-weighted grade points by the total enrolled credits, and the complete institutional grade sheet along with class performance statistics is rendered dynamically on both the Web Dashboard and the Command Prompt Console.

RESULT:

Thus, the Student Academic and Continuous Evaluation System for Kongu Engineering College (Autonomous) Regulation 2024 was successfully designed, developed, and tested using Java (JDBC) and MySQL. The system accurately automates the calculation of Continuous Internal Assessments (CIA), End Semester Examinations (ESE), Letter Grades, Attendance Eligibility, and Semester Grade Point Average (SGPA).